

# Necromancing Diels: computerising the phonological analysis of early Slavonic texts using existing treebank data and a Late Common Slavonic computerised inflectional morphology

## 0. Introduction

Much progress has been made in the last twenty years in early Slavonic corpus linguistics as a result of the Old Church Slavonic part of the PROIEL project (Haug & Jøhndal 2008) and its subsequent expansion as the TOROT treebank (Eckhoff & Berdičevskis 2015), such that currently just over 240,000 words of canonical OCS have been manually lemmatised, part-of-speech and morphologically-tagged, and syntactically parsed. The focus of these projects, however, has been exclusively on the higher-level linguistic domains of syntax, semantics, and pragmatics: surface-morphology has been of only incidental concern, for example in investigations into differential-object marking (Eckhoff 2015, 2022), but the sort of inflection-class-marking which would enable the retrieval of, say, masculine o-stems vs i-stems is lacking. The needs of historical phonologists especially are ill-served, **since some of the texts (esp. Kiev Folio, Codex Suprasliensis, and partially Codex Zographensis) contain quite severe typographical inconsistencies and errors that make them dangerous to use without reference to the manuscripts.**

That being said, enough information is included in Eckhoff's lemmatisation and morphology-tagging that, with a few exceptions (e.g. comparatives), the morphological shape of the inflected text-forms can be predicted from just the tag-information, provided that inflection-class annotations are added to the lemmas. This means that the immediate Late Common Slavonic ancestors of surface-text forms can be generated by reconstructing and inflection-class-marking the LCS lemmas, and then using a computerised LCS inflectional-morphology to inflect each text-word's lemma according to its morphology-tag<sup>1</sup>. Such LCS reconstructions are an extremely useful form of 'phonological annotation', because theoretically all the information required to give rise to an attested form must by definition be present in any correct reconstructed proto-form, and the complete regularity of the idealised LCS forms makes texts predictably searchable regardless of orthographic variability, abbreviations, or other irregularities in the surface-texts. When applied to whole texts, they make the exhaustive investigation of almost any phonological or orthographic question trivially easy compared to manually reading and extracting relevant forms, or using TOROT's existing lemmatisation and morphology-tagging to try to gather morphological categories which might contain the sound-groups one is interested in.

The goal of this article is to describe just such a computerised LCS inflectional-morphology, and to show how it can be used to "autoreconstruct" different OCS texts from the TOROT corpus, while explaining how difficulties caused by things like morphological innovations, badly-integrated foreign loanwords, or insufficiently-precise tagging-data can be overcome. The resulting 'phonologically annotated' texts should allow investigators of the lower-level linguistic domains (phonology, orthography, and morphology) to benefit from the huge amount of manual annotation work that has gone into the OCS part of TOROT, and the computerised LCS inflectional-morphology by itself is valuable in that it allows linguistically-rigorous and comprehensive inflection and conjugation-tables to be generated<sup>2</sup>.

1 Morphological innovations and variations are detected by inspecting the text-forms and then applying 'alternative' endings as specified in the inflectional-endings database; see Section 2.3.

2 The lemma and inflection-class data included as a supplement to Поливанова's (2013) comprehensive OCS grammar, <https://integral.github.io/osd/>, would in principle allow similar tables to be generated, but as far as I know Поливанова has not digitised the actual endings, nor written any scripts to generate paradigms. Additionally, because she bases her forms on a normalised OCS, rather than LCS, phonemic system, a massive amount of information has been needlessly destroyed (to give just one obvious example, the syllabic-liquids \*ř \*ľ \*ṛ \*ḷ are represented by their OCS reflexes /рь лъ рь лъ/, meaning there's no distinction between the syllable-rimes of roots like /-срьд-/ <\*sřd/ and /-крьст-/ <\*krbst/). An inflectional-morphology based on her data would therefore only be good for generating the (in reality unattested and fictional) "normalised" form of OCS, whereas my LCS forms are trivially convertible into *any* dialect of OCS, as well as pre-Jer Shift Old East Slavic (see pg. 17 and Figure 7), because they represent the language at a stage before any such dialect-specific phonemic mergers.

A web-interface for autoreconstructed TOROT OCS text is being developed at <https://ocstexts.co.uk>, which aims to demonstrate the benefits of this extra layer of annotation by offering things like LCS phoneme-search of texts, conversion of each autoreconstructed word into canonical "normalised" OCS, for easy comparison with the manuscript-form, and computer-generated inflection and conjugation-tables for every reconstructed lemma. It also includes TOROT's existing morphology, lemmatisation, and (for three of the texts) Greek source-alignment data, plus integration of the recently digitised OCS dictionaries hosted at [gorazd.org](http://gorazd.org).

Slavonicists are in the fortunate position of having not only a widely and diversely attested set of daughter-dialects from which to reconstruct the proto-language, but also a historical written record that is so close in time to the Common Slavonic period that the far-southeastern Bulgarian dialect of Thessaloniki was considered a suitable vehicle for proselytising the proto-Czech speakers of Moravia. The hypothetical LCS system which I employ for my reconstructions is therefore mostly uncontroversial and widely agreed-upon (except in trivial matters of notation); nevertheless, a corpus-focused project like mine where the aim is to reconstruct *all* words of a text requires an exhaustiveness and rigour that leaves no room for the dodging of difficult points, so Section 1 of this article is taken up with a comprehensive exposition and justification of my chosen LCS system. Section 2 then looks in more detail at my computerised LCS inflectional-morphology and the structure of the existing TOROT data, and shows that together they can be used to produce the direct phonemic ancestor-forms of text-words, despite various challenges, including the morphological variation and restructuring which is occurring in even the earliest texts. [Section 3 ...](#)

## 1. My Late Common Slavonic (LCS) phoneme-system

The premise of my chosen form of "phonological annotation" is that the earliest Slavic texts reflect languages which are **structurally** close enough to the broadly-agreed-upon system of Late Common Slavonic that the forms underlying the manuscript-spellings are more or less trivially derivable (by the application of sound-change rules) from their theoretical LCS ancestors. By 'structurally' I am referring to structure at the phonological level; structural changes at higher levels of analysis (i.e. inflectional morphology, derivational morphology) are of no concern unless they are **made possible only by intervening phonological changes**.

My contention is that before about 1100 not enough of these structural changes are in evidence in any Slavic text, and thus texts can be relatively straightforwardly indexed using a well-chosen LCS system. Before giving examples of structural changes that are problematic for such an indexing-system, it's necessary to first lay out my LCS system in full:

In order to account for as much of the subsequently attested Slavic as possible, a point after the monophthongisation of diphthongs, but before the Second and Third Velar Palatalisations (PV2 and PV3) is chosen as the point of departure, because of the difference between the West Slavic /š/ and South/East /ś/ reflex of these two palatalisations of \*x (Cz. loc. pl. *dušič* vs Suprasliensis *доуѣхъ* <\*duxěxъ; Polish *wszak* vs Supr. *вѣакъ*, Ru. *всяк[уй]* <\*vьx-akъ), as well as the probable complete absence of PV2 in northern East Slavic<sup>3</sup> (Old Novgorodian, see Zaliznjak 2004: 42-45 for

3 The evidence regarding the possible absence of PV3 from Novgorodian is far less convincing: the Birchbark letters abound with examples of the PV3 reflex of \*k (e.g. letter №439 from around 1200 has *свннѣѣ* <\*svinьkь and *полотѣнѣѣ* <\*polьtьnьka), and those of \*g are not unknown: Zaliznjak (2004: 47) admits that palatalised forms of the Germanic loan *кѣнѣз-* <\*kьneg- are the rule, but considers this to be a "supradialectal" word originating outside of the Novgorodian dialect-area; Галинская (2014: 10) is less convinced and adduces the form *оуѣрѣзн* (cf. Russian *серѣга*, commonly assumed to be an Oghur, i.e. Bulgar, Turkic loan, cognate with e.g. Kazakh *сырға*) 'earrings' from letter №429 as a word of "вполне бытового характера" which thus supposedly shows a native Novgorodian reflex of PV3 of \*g. More importantly, as Галинская (op. cit.) points out, in all of the well-known Novgorodian forms of the pronoun \*vьxъ 'all' which supposedly show a lack of PV3 by retaining both /x/ and back/hard desinences (e.g. fem. gen. sg. *вѣхѣ* <\*vьxoje from letter №850), and which come from letters which otherwise correctly convey the jers (by

the evidence), and the blocking of PV2 by an intervening \*v in West Slavic (Pol. *gwiazda*, Cz. *květ* <\*gvězda, \*květъ, etc.).

To be explicit, the native phonemes in my LCS system are given in the tables below:

Table 2: LCS Vowels after the monophthongisation of diphthongs

|      | Front  |    | Back      |   |
|------|--------|----|-----------|---|
| High | i      |    | y         | u |
|      | ĭ ѣ ѣ̇ |    | ŷ ɣ ѣ̇ ѣ̇ |   |
| Mid  | e      | ę  | o         | o |
|      | ě      | ě̇ |           |   |
| Low  | Æ      |    |           |   |
|      |        | a  |           |   |

Table 1: LCS consonants before PV2/PV3 (adapted from Winslow 2022: 304)

| Labial |   | Dental |   | Palatal |   | Velar |   |
|--------|---|--------|---|---------|---|-------|---|
| m      |   | n      |   | ń       |   |       |   |
| b      | p | t      | d | ħ       | ḥ | k     | g |
|        |   | s      | z | š       | ž | x     |   |
|        |   |        |   | č       |   |       |   |
|        |   | l      |   | ĺ       |   |       |   |
|        |   | r      |   | ř       |   |       |   |
| v      |   |        |   | j       |   |       |   |

### 1.1 Foreign sounds

In addition, the following symbols are used for phonemes of wholly foreign origin in order to represent badly-integrated foreign borrowings, whose level of integration into the native system we deliberately do not take a position on: /k ǵ x f ü/, e.g. in respectively *кѣтъ* <\*kĭtъ, *ѣмѣмонъ* <\*ĭǵemōnъ, *хитѣмонъ* <\*xĭtōnъ, *иосифъ* <\*ijosifъ<sup>4</sup>, and *мѣро* <\*mĭro. Almost none of the words containing these symbols would actually have existed in the language during Common Slavonic times, but they need to be included in the indexing-system because they often contain native Slavic elements (f.ex. inflectional endings). Normally they represent specific sounds in the source-language (usually Greek), so including them is useful for investigating the process of these sounds' integration into the native systems. For instance, the extent to which Greek /ü/ is integrated into either native /i/ or /u/ can be seen in variations in the OCS spellings of the word for 'Egypt' (\*ēǵŭpъtъ): *ѣǵŭpъtъ* {ēǵŭpta} vs *ѣǵŭpъtъ* {ēǵŭpta} vs *ѣǵŭpъtъ* {ēǵŭptŭ} vs *ѣǵŭpъtъ* vs *ѣǵŭpъtъ* {ēǵŭpta}<sup>5</sup>, etc.. One might also ask whether a separate <ѣ> letter for /ǵ/ (and the writing of <ѣ>/<ѣ> with the palatalisation-diacritic) could be linked to the inadmissibility in the native systems of soft

writing <ѣ, ѣ> for \*ѣ and <ѣ, ѣ> for \*ѣ, the weak-je is always written with <ѣ, ѣ>, unambiguously suggesting a /ѣ/ pronunciation. These forms therefore more likely point to a LCS doublet-form \*vѣxъ which would never contain the conditioning environment for PV3 anyway, and thus you can't use them as evidence of a lack of PV3 in Novgorodian (on the plausibility of such a doublet see Галинская (2014: 14), though cf. Zaliznjak's (2004: 54) less convincing explanation of the /ѣ/ in these words as an assimilation of original /ѣ/ to the back-vowels of the following syllable).

4 Of course the sequence /jo/ violates LCS phonotactics as well.

5 Forms are given as they appear in the manuscripts; modern fonts and Unicode symbols mean that the misleading and unhelpful practice of transcribing Glagolitic into Cyrillic is no longer defensible. Latin-based transliterations of Glagolitic forms are given in {curly braces} according to the table in Appendix 1, but it should be emphasised that no transliteration system can adequately convey the internal logic of the Glagolitic, and those who study OCS through transliterations are hamstringing themselves. Where specific forms from Eckhoff's corpus-texts are cited, they are hyperlinked to the corresponding part of the text as it appears on the [ocstexts.co.uk](http://ocstexts.co.uk) website. The Glagolitic texts there are unfortunately still Cyrillicised and should be used with care, particularly ones like Psalterium Sinaiticum where certain ruesome decisions from the editors Severjanov (1922) and Mareš (1997) have been compounded by further information-destroying simplifications (for instance, Severjanov transliterates <ѣ> with <ѣ>, but Eckhoff then replaces Severjanov's roundy diacritic with a tilde, leading to nonsense transcriptions like Psalm 8 <ѣѣѣѣ> for ms. <ѣѣѣѣ>). Cleaned up digitisations, using the Glagolitic originals as the base-text, and a mechanism for displaying manuscript-images, are planned for the near future.

6 Other troublesome pre-LCS morphological isoglosses reflected in the texts include the masc./nt. instr. sg. \*o- and \*jo-stem endings \*-ѣтъ/\*-ѣтъ (N.Sl., e.g. KF ѡбразѣтъ {obrazŭmĭ}, Uspensk. Sbor. кнѡзѣтъ) and \*-омъ/\*-емъ (S.Sl., e.g. Supr. ѡбразѡмъ, кнѡзѣмъ), which are most commonly (e.g. Olander 2015: 168) thought to be analogical replacements of the original instr. sg. ending ECS \*-ā which is preserved in the adverb \*vŭčera ‘yesterday’; and the \*-тъ (N.Sl.) vs \*-тъ (S.Sl.) verbal endings of 3<sup>rd</sup> sg. and pl. present (plus its extension to 2<sup>nd</sup> and 3<sup>rd</sup> sg. aorists like OCS научѣтъ, OR (Uspensk. Sbor.) бѣсѣтъ, научѣтъ). Here I have no choice but to index them with dummy-symbols in the database: \*-Omъ/\*-Emъ for the instr. sg. ending and \*-tQ for the verb-endings.

which the Glagolitic alphabet was devised) for the reflexes of LCS \*č/š/ž/h + \*Ē (e.g.  $\text{обѣцѣ}$  {oběcělŭ} <\*ob-věhĒlŭ,  $\text{дусѣми}$  {dušēmī} <\*dušĒmi), as well as occasional traces of such spellings in later Glagolitic OCS (e.g. Psal.  $\text{чѣсѣ}$  {čěšě} <\*čĒšě); and

all together point very strongly towards there having occurred a split on the Southeastern periphery of Slavic between LCS dialects which have /ě/ < \*Ē and the majority of the rest which got /a/, and that original OCS ('Urkirchenslavisch') was an \*Ē > /ě/ dialect. I posit that \*Ē remained until the opposition /a:/ě/ after palatal consonants was reintroduced when PV2 and PV3 brought new soft-consonants /c š dž/ into the system, which could be followed by both /a/ and /ě/: fem. nom. sg. /stǫdža/ < \*stǫga (PV3) vs fem. loc. sg. /nodže/ < \*nogě (PV2) (Winslow 2022: 304-305). Thus since my LCS system is based on a point just *before* PV2 and PV3, I must also retain the \*Ē archiphoneme.<sup>7</sup>

7 It's possible to argue that the short *\*Ē* counterpart to *\*Ē* persisted in East Slavic until after the Fall of the Jers, and that the ESL, so-called e > o shift before hard-consonants / back-vowelled syllables is actually just the resolution of this archiphoneme as /o/ (where palatalisation of the preceding consonant remained, in e.g. Ukr. *бджола* <\*bĥĒla, or was newly phonemicised, in e.g. Ru. *вѣсла* <\*v'Ēsla <\*vesla), and that there was never a stage when these words had /e/ (based among other things on <o> spellings regardless of stress after palatal-letters in very early texts, and even after the letters for secondarily-soft LCS plain consonants in the Birchbark documents (Le Feuvre 1993, Nakonečnyj 1962), but there isn't space to elaborate on the issue here (see Winslow 2022: 304 fn.16). Unlike the situation with long *\*Ē*, OCS shows no sign of anything but an /e/ reflex of short *\*Ē* (and indeed the fact that the East Slavs inherited their writing system ultimately from the Urkirchenslavisch system designed for such a dialect, rather than one which had a clear way of writing /soft consonant/ + /o/, is likely the reason that /o/ reflexes are so rarely detectable in the early texts, since <e> had to be used for both /e/ and /'o/, cf. the spelling *ѡвѡшанъ* of the Kipchak word /jovʂan/ 'wormwood' in the Hypatian Codex, whose modern cognates (Turkmen *ýowʂan* /jowʂan/, Kazakh *жуған* /žuwsan/, Azeri *yovʂan*) unambiguously point to a Kipchak /o/), and the history of the East Slavic /o/ reflexes remains the subject of much disagreement, so it's simpler for everyone if I continue the traditional practice of writing LCS *\*e* after palatals, even if that strictly speaking is inconsistent with my use of *\*Ē*.



ʋḷik-/ʋḷic- {vľik-/vľic-} and all 15 instances of its \*-mľič- root with -ḿḷič- { -mľič- } (e.g. ḿḷičašn {umlīčašn}). Therefore, taken as a whole the Slavic evidence pretty securely points to front and back variants of both syllabic liquids, and for searching purposes it's far preferable to denote them with separate symbols<sup>8</sup> rather than as the sequences /ɛr ɛr ɛl ɛl/<sup>9</sup>.

### 1.3 Consonants - Dejotation

Reflexes of the so-called jot-palatalisation are all written either as unitary palatal phonemes, or in the case of jot-palatalised labials as /vľ mľ bľ pľ/, rather than as sequences of consonant + /j/, hence /ń ľ ř/ for \*nj \*lj \*rj. The 'dejotated' reflexes of \*tj (and \*kt+front-vowel) and \*dj are denoted using the modern Serbian Cyrillic letters /h/ and /j/ respectively, because the commonly used alternatives, i.e. /t̚ d̚/ (as used in e.g. Olander 2015) or /k̚ g̚/ (as used by me in Winslow 2022), or variations thereof, are visually too close to symbols used elsewhere in the system. /k̚, g̚/ are anyway already used in my system for foreign /k, g/ before front-vowels, and /t̚ d̚/ look too similar to the common denotations of secondarily-palatalised post-Jer Shift /t' d'/, as used in discussions of systems like Russian or Eastern Bulgarian where they arise.

The compelling hypothesis, first proposed by Durnovo (1929: 55-58) but most recently elaborated by Vermeer (2014: 209-214), and accepted by Mathiesen (2014: 197 fn. 22) and Winslow (2022: 310 fn.25), according to which the Urkirchenslavisch reflexes of \*h, ħ were close enough to foreign /g k/ before front-vowels that the original Glagolitic system used <Ѡ ѡ> for both sets of sounds (i.e. alongside attested ḡḷemōnŭ {igemonŭ} < ἰγεμων would have been \*\*ḡḷemōnŭ {osq̇eni} < \*osq̇eni, and alongside attested dŭčerĭ {dŭčerĭ} < \*dŭčerĭ would have been \*\*dŭčerĭ {cĭnŭsŭ} < κῆνσος<sup>10</sup>), does not prevent us from keeping the foreign sounds separate for our LCS stage, since clearly they differed enough in all the dialects underlying actually attested OCS to be written separately.

Pre-dejotation \*stj and \*zdj are differentiated from the PV1 reflexes of \*sk and \*zg by writing the former as \*šh and \*žh and the latter as \*šč and \*žž, even though their modern reflexes do not differ from each other anywhere and so must've fallen together in the CS period, because they often alternate with their respective un-palatalised counterparts morphologically and derivationally, e.g. očistiti: očišhenje vs. jьskati: jьščq, jĀzdi: jĀžh vs jьzgъnati: jьžženq.

There are convincing arguments for PV2/3 having preceded dejotation, at least in more central areas, most recently presented in e.g. Vermeer (2014: 197) and Wandl & Kavitskaya (2023: 244-247), and therefore it could be objected that my system, which contains the dejotation reflexes /h ħ ħ ľ ř/ but not the PV2/3 reflexes /c ś dž/, is ahistorical. However, it should be reemphasised that the

8 In the database I will have to use the single Unicode characters <ř ľ ľ ħ>, rather than what's shown in my table, since the latter cannot actually be rendered without using the letters for /r ľ ľ ħ/ plus the 'combining ring below' U+0325 symbol, which means searches for the consonantal liquids on their own will also return results containing syllabic liquids. A similar problem affects /ġ y/, which I will have to replace with <ġ ŷ>.

9 To my mind the only evidence in support of a genuine jer + liquid stage comes from the paradigms of verbs like OCS сѣръти < \*sĕrtĭ, where the syllabic /ř/ in the stem alternates with /ɛr/ depending on the vocalicity of the following morpheme: the e.g. 3sg. pres. \*sĕrĕtĕ (Zogr., Supr. сѣръте) or (one possibility of the) 3<sup>rd</sup> sg. aorist \*sĕrĕ (Supr. сѣръе) must have /ɛr/, while the 3<sup>rd</sup> pl. aorist \*sĕrtĕ (Supr. сѣръша) and the other possibility for the 3<sup>rd</sup> sg. aorist \*sĕrtĭ (Psal. сѣръ {sŭrtĭ}, or with a different prefix Mar. сѣръ {otrŭ} < \*otrĭ), being word-final or pre-consonantal, must be syllabic /ř/. The same alternation occurs in the zero-grade forms of verbs like \*umerti, as is clear from the Polish reflexes *umarĭ* < \*umřĭ vs *umrę* < \*umĕrĕ. The argument could be that at some stage, before the LCS tendency towards Open Syllables became dominant, the stems in these paradigms were surely unitary /tĭr/, /mĭr/, i.e. 3sg. aor. /sŭ.tĭ.re/ vs 3<sup>rd</sup> pl. aor. /sŭ.tĭr.še/, and that the latter's closed /tĭr/ syllable was only forced to open itself up by changing to /tĭ/ because of the Law of Open Syllables. Thus at least one source of the syllabic-liquids could be shown to have developed from a vowel + liquid stage, but that still doesn't prove that they all did, or that the change of /tĭr/ to /tĭ/ in these verb-forms was not merely a move to an already-existing syllabic-liquid phoneme.

10 Interestingly, this aspect of the hypothesised Urksl. orthographic system has rearisen in the modern Macedonian standard due to Turkish loanwords: *кемер* < Tk. *kemer* 'belt', *ке* < \*[x̥]he[t̚]; *ѓон* < Tk. *gön* 'leather', *меѓу* < \*meĭu.

primary goal of my LCS reconstructions is to act as an index which allows reflexes in texts to be found, not to be a historically realistic description of some actually-existing LCS dialect. The absence of PV2 in Novgorodian shows that it can't have preceded dejotation everywhere in Slavic, and in any case the replacement of the sequences /tj dj nj lj rj/ by articulatorily distinct combined units, no longer associated by speakers with their /t/ and /j/ phonemes, is structurally completely irrelevant unless and until these new units merge with existing phonemes (or new sequences of dental + /j/ are introduced), as e.g. in the KF dialect where /tj/ merged with /c/ from PV2/3, or in ESL where it merged with /č/ from PV1. A language which had distinct Czech-like palatal [c, ʝ] reflexes of \*tj and \*dj, and also no new sequences of [tj, dj], could not convincingly be argued to have undergone dejotation at the phonemic level, as these new units would just be phonetic realisations of /tj, dj/. Analysed like that, the symbols /ħ ħ́ ħ́́ ħ́́́/ in my system strictly speaking would really just be cover-symbols for the pre-jotation sequences, but such notation is preferable since it prevents searches for groups containing /j/ alone from returning results polluted by all the dejotation-groups. As I explored in my previous article (Winslow 2022), the status of /j/ as a phoneme in the earliest OCS texts is an intricate problem, so the ability to investigate the reflexes of \*j in isolation from the dejotation-reflexes is important.

#### 1.4.1 Word-initial \*jĀ-/a-

Forms with insecure etymologies can't under any methodology be used as good evidence in phonological investigations, so in difficult cases like the above I simply mark the lemma in the database and provide some short discussion, so that eventually the web-interface can flag such forms in some way and inform users of the specific difficulties.

### 1.4.2 Jers before \*j

<\*osŋdětъ ѱъ; KF *мѣстѣмъ* {milostĭjŋ} vs *мѣстѣмъ* {milostĭjŋ} <\*milostĭjŋ, *вѣсѣмѣмъ* {vĭsemogŭi} vs *вѣсѣмѣмъ* {vĭsemogŭi} <\*vĕxomogy-ъ). Of importance here is the fact that the same orthographic system characterises both pre- (i.e. KF) and post-Jer Shift texts; that even in strong position in a text like Zographensis, which shows pretty clear signs of having undergone the Jer Shift, spellings like *ударѣи* {udarĭi}, *болѣи* {bolĭi}, *вѣстѣи* {vnstĭi} for what in the live dialect underlying Zogr. must surely have been /udarĭj/, /bolĭj/, /vęstĭj/, are not infrequent<sup>11</sup>. The fact that the same alternation occurs in the pre-Jer Shift KF (i-stem gen. plurals *зповѣди* {zapovędĭi} <\*zapovędĭjъ vs *людѣи* {lŭdĭi} <\*lŭdĭjъ) suggests that it is a common inheritance from the Urkirchenslavisch spelling system, and thus that in pre-Jer Shift Slavic the difference between /ъ ѱ/ and /i y/ was neutralised before /j/, and we should perhaps posit archiphonemes (which I call /Ĭ/ and /Ŷ/) in this position. These archiphonemes are, in slightly different terms, effectively posited by Trubetzkoy (1954: 70) in his analysis of the Urkirchenslavisch phoneme-system<sup>12</sup>.

Difficulties arise though when deciding how to denote foreign sources of /ij/<sup>13</sup> which may or may not have been integrated into the native system as reflexes of /ĭj/: words like **мариѡ** < Μαρία, **стаѡи** < στάδιον, which are well-integrated into the morphological system as a fem. ja-stem and masc. jo-stem respectively, could either be reconstructed as consciously-foreign \*marijĕ, \*stadijĕ, or as nativised \*marĭjĕ, \*stadĭjĕ, but there are no occurrences of jer-spellings in these words in the OCS texts in TOROT. Other similarly-Greek words like **дѡѡволѡ** (< διάβολος), however, do show up in OCS with jer-spellings: Supr. **дѡѡвола**, Zogr. Luke 8 and Psal. Psalm 108 **дѡѡвола** {dĭěvolŭ}, which (alongside the modern Macedonian *ѓавол* with the reflex of \*ĥ produced by the Macedonian so-called ‘new jotation’ of /d/ after the fallen jer brought it into contact with /j/) clearly suggest an early adaptation of this foreign /ij/-group to native /ĭj/. Old Russian texts even show spellings of **мариѡ** suggestive of full nativisation: Laurentian Primary Chronicle **мѡрьѡ**, **мѡрьѡ**, Zadonshchina **мариѡ**, **мариѡ**, as well as First Novgorod Chronicle gen. sg. **васнѡѡ** (jo-stem **васнѡѡ** < Βασίλειος).

### 1.4.3 Word-initial \*j<sub>B</sub>-/\*ji-/\*i-

11 Marianus and Psalterium Sinaiticum, on the other hand, frequently show a Russian-style /ej/ reflex of strong tense \*ɤj: Psal. ᚱᚱᚱᚱᚱᚱ {vrabei} <\*vorɤɤj, ᚱᚱᚱᚱᚱᚱ {plūtei} <\*plɤtɤj, ᚱᚱᚱᚱᚱᚱ {mīnei} <\*mɤnɤj; Mar. ᚱᚱᚱᚱᚱᚱᚱᚱ {zapovědei} <\*zapovɤɤj, ᚱᚱᚱᚱᚱᚱ {udarei} <\*udaɤɤj, ᚱᚱᚱᚱᚱᚱᚱᚱᚱᚱ {gvozdeinūj} <\*gvozɤɤjny-jɤ. Psal. (Psalm 21) even has a past act. part. Nsg. masc. def. form ᚱᚱᚱᚱᚱᚱᚱᚱ {strūgoi} <\*jɤstrgɤj that suggests an /oj/ reflex of \*ɤj.



I make an exception for certain forms of the personal-pronoun \*jъ, however, and write \*jimъ, \*jima, \*jixъ \*jimъ and \*jimi for the masc/nt. instr. sg. and dat./instr. dual/pl., because Czech here has *jim jich jimi*.

Prefixed forms like \*do-jyti 'to come, arrive' for morphological reasons have to be distinguished from the class 4 verb \*dojiti/dojiši/dojimъ etc. 'to breastfeed' (and its derived noun \*dojidlika), a difference which is reflected in the modern Ukrainian *doiťmu* (<\*dojъti with compensatorily-lengthened /o/ > /i/) vs *doimū*. Thus /i/ can follow /j/ when the former is part of a morpheme which just happens to be stuck onto a /j/-ending stem: I similarly allow words like \*šujika (шуйица) and \*vojinyъ 'warrior' (воинъ, as opposed to \*vojъnyъ, the gen. pl. of \*vojъna), or the loc. sg./pl. desinences of any jo-stem noun whose stem ends on /j/, e.g. Psal. *жрѣбїи* {žrěbūi} <\*žerbyji.

No Glagolitic text makes any effort to distinguish /je/ (after vowels or word-initially) from post-consonantal /e/, writing both with <ǣ>, unlike the situation with the reflexes of \*ję vs \*ę, where in Zogr. and Mar. and partially in Assem. (Белчева 1981: 168) the full front-nasal digraph <ǣę> is reserved for \*ję, while just the second 'nasalising component' <ę> is used for post-consontal \*ę, e.g. Mar. 3<sup>rd</sup> pl. aorist **ꙗꙋꙋꙋ** {ęsn} <\*jęse, as opposed to KF **ꙗꙋꙋꙋꙋ** {prięti} <\*prięti vs **ꙋꙋꙋꙋꙋꙋ** {vüželi} <\*vǣželi<sup>15</sup>.

14 Spellings like Zogr. Mark 13:3 “*ᲡᲉᲗᲣᲹᲗ. ᲙᲁᲑᲗᲣᲹᲗ. ᲙᲁᲑᲗᲣᲹᲗ.*” {petrū. i ěkovū. i oannū} “Peter and Jacob and John” would suggest that this initial \*i- can get dropped after an /i/ of a preceding word, but whether this points to a dropping of the non-native \*i-, simple deletion of a double /i i/ (haplology), or a native-like reflex of a weak-jer /\*i jəj.Ěkovŭ/ > /i jakov/, is not really knowable, so indexing such words with a markedly foreign initial \*ij- group is again the best way of allowing such difficult cases to be investigated.

[illegible]

16 The leftover 14 are things like 1st. pres. dual. **ймавѣ** which Eckhoff's corpus wrongly lemmatises as **имати** instead of **ймѣти**, and which thus get reconstructed as \*jěmǎvē instead of \*jǎmavě. At the time of writing only 3227/6862 Suprasliensis lemmas have been reconstructed, but those 3227 cover 89713/99194, or 90.4%, of the words.

Therefore even though \*sr > \*str and \*zr > \*zdr appear to be simply voiced and unvoiced variants of the same sound change, the practical effects are very different because the former is, from the LCS perspective, totally ‘opaque’, since it only occurs in roots and thus is not analysable by

For this reason I don't include /zdr/ <\*zr at prefix or preposition-boundaries in my LCS system, so that investigators can see for themselves the extent of each text's adherence to the expected phonological development vs restoration of /zr/ under morphological pressure.

### 1.6 Morphological innovations that scupper LCS reconstruction

An example of such morphological change contingent upon structural phonological change, leading to forms which preclude any direct LCS-stage reconstruction, is the replacement of i-stem endings with those of the corresponding jo- or jā-stems, in nouns whose stems end on labials or the subset of LCS dental consonants which lack palatal counterparts, viz. /d t s z/<sup>19</sup>. Evidence for such a change is furnished by the Old Russian masc gen./acc. form **ТАТА** from the 1229 Treaty between Smolensk, Riga and Gotland (Version A). LCS \*tatъ is a masc. i-stem noun with genitive \*tati, as it still appears in the Suprasliensis translation of John Chrysostom's Homily for Holy Thursday (...то кажетъ владѣицы ѹловѣколюбїе: іако прѣданника развоѣника **ТАТИ**...), but in the dialect underlying the 1229 Treaty the rise of phonemically palatalised /t'/ after the Jer Shift means that the stem (and the nom. sg. **ТАТЬ** /tat'/) of this noun now ends on the same class of "soft" consonants as original jo-stem nouns like \*koń > /kon'/, where the original LCS palatal \*ń has fallen together with

19 In some dialects (notably East Slavic) the PV3 reflexes \*š and \*ž became palatalised counterparts to plain /s z/, i.e. /s' z'/, and merged with the /s' z'/ that developed from LCS \*s,z before front-vowels, but in most OCS they seem to have just hardened to /s, z/: searching my database for the sequence \*ьхѡ, for example, turns up exclusively <ѡѡ> spellings in Marianus, just one <ѡѡ> in Zogr., and exclusively <ѡѡ> in Suprasliensis, with only Assem. and Psal. containing a significant minority of <ѡѡ> spellings.

LCS /Ē/, though, by definition can only occur after LCS palatal consonants (see above), so a reconstruction \*tatĒ is just nonsensical. In the case of the dat. sg. /u/-desinence (which isn't attested in our Treaty but exists in modern Russian *мамю*), we don't even have an LCS archiphoneme available to signal a preceding soft-consonant; there's simply no way of getting from LCS \*tatu to Russian /tat'u/, because such a form was only made possible by the rise of phonemic /t'/, so our ability to index it with our LCS system is gone.

Forms like **T A T A**, then, though they frustrate our goal of reconstructing entire texts, do provide us some objective measure of ‘linguistic distance’ between stages of a language, because their existence presupposes at least one intervening stage where the structure of the phonological system has changed enough from our LCS stage to have caused/allowed restructuring of the morphological system.

## 2.1 The existing TOROT data

20 Russian feminine i-stems like *весь* (<\**вьсь*, ‘village’) do not fall together with ja-stems in the way the masculines like \**zvěť* fall together with jo-stems, but they do all still take the /am, ax, ami/ endings in the dat., loc. and instr. pl., e.g. *весьям*, which contain the same LCS ja-stem \*-*Ē*- vocalism which can only occur after LCS palatals, meaning they too end up totally unreconstructable due to an illegal \*\**s*Ē sequence.

21 Numerous spellings in Supr. like *ѡура* <\*buřĀ, *ѡукарѣтъ* <\*ukařĀjetŭ, *моуру* <\*mořu etc., however, point to a hardening of LCS palatal \*ř to plain /r/, so it's difficult to know whether *ѡѡрѣтъ* spellings stem from a genuine /zvěra/ form in the history of the language, or if they instead represent synchronic /zvěra/, i.e. with a hard o-stem ending, but with confusion by the scribe between <ра> and <ря>/<ръ> spellings for what in his/her dialect would've all been /ra/.

22 Except the numerous gen. sg. *ꙗꙗꙗꙗ* and dat. sg. *ꙗꙗꙗꙗꙗ* for the i-stem \**gospodъ*, but this word seems to be an isolated special case, because it bafflingly turns up even in early Glagolitic OCS with endings like *ꙗꙗꙗꙗ* {*gŭi*} (ju-stem dat. sg. \*-*evi*, cf. Supr. *ꙗꙗꙗꙗꙗꙗ*), *ꙗꙗꙗꙗ* {*gŭ*} (jo-stem dat. sg. \*-*u*), and *ꙗꙗꙗꙗ* {*gě*} (jo-stem gen. sg. \*-*Ē*). See Van Wijk (1929).

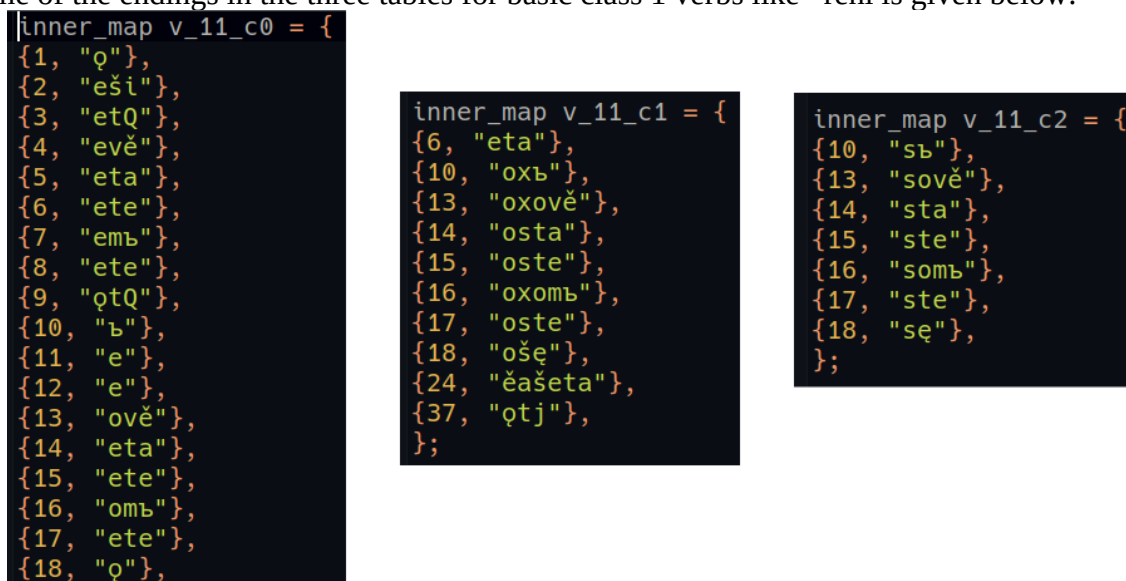




The Autoreconstructor reads the morphology-tag character-by-character and numerifies each field, and from that it computes a number which corresponds to the row of the inflection-table where the endings for that particular word's inflection-class are stored. For verbs this will be a number between 1 and 44 (9 person/number combinations for each of present, aorist, imperfect, and imperative, plus 8 non-finite forms,  $9 \cdot 4 + 8$ ), and for nominals between 1 and 63 (7 cases \* 3 numbers \* 3 genders). A version of the function which actually implements this tag-reading process can be seen in the `OcsServer::numerifyMorphTag()` function here:

[https://github.com/12401453/ocs\\_server/blob/main/OcsServer.cpp#L2714](https://github.com/12401453/ocs_server/blob/main/OcsServer.cpp#L2714).<sup>25</sup>

Currently each inflection-class has up to three tables associated with it, the first of which is full (i.e. contains 44 or 63 entries) and holds the 'basic' or 'correct' (from the LCS perspective) endings. The second and third tables are 'sparse', in that they only contain entries for those parts of the paradigm where we expect to encounter alternative forms, with those I consider 'deviant' or 'innovated' held in table 2, and 'alternative' but still 'correct' endings (i.e. LCS allomorphs) in table 3. An example of some of the endings in the three tables for basic class 1 verbs like *\*rehi* is given below:



```
inner_map v_11_c0 = {
{1, "q"},
{2, "eši"},
{3, "etQ"},
{4, "evě"},
{5, "eta"},
{6, "ete"},
{7, "emъ"},
{8, "ete"},
{9, "qtQ"},
{10, "ъ"},
{11, "e"},
{12, "e"},
{13, "ově"},
{14, "eta"},
{15, "ete"},
{16, "omъ"},
{17, "ete"},
{18, "q"},
};

inner_map v_11_c1 = {
{6, "eta"},
{10, "oxъ"},
{13, "oxově"},
{14, "osta"},
{15, "oste"},
{16, "oxomъ"},
{17, "oste"},
{18, "ošę"},
{24, "ěašeta"},
{37, "qtj"},
};

inner_map v_11_c2 = {
{10, "sъ"},
{13, "sově"},
{14, "sta"},
{15, "ste"},
{16, "somъ"},
{17, "ste"},
{18, "se"},
};
```

Figure 2: Inflection-endings for class 1 verbs like *\*rehi*, *\*greti* stored using C++ `std::unordered_map<int, std::string>`

Here the 'deviant' endings consist mostly of the 'extended S-', or \*-ox- aorists (e.g. Assem. *ꙗꙗꙗꙗꙗꙗꙗꙗꙗ* {narekošn}), while the third 'alternative' table contains the primary S-aorist endings (e.g. Assem. *ꙗꙗꙗꙗꙗꙗꙗꙗꙗ* {pogrěsn} <\*pogrěb-se). Clearly those primary S-aorist endings just being stuck onto a stem like *\*greb-* would not produce the correct LCS form, so for certain inflection-classes there are post-processing functions which do things like replace all "ebse" with "ěse", or (for stems which end on RUKI-consonants like *\*rek-*) all "eks" with "ěx". The full routines for verbs can be found in [https://github.com/12401453/ocs\\_server/blob/main/autoreconstructor/verb\\_cleaning.h](https://github.com/12401453/ocs_server/blob/main/autoreconstructor/verb_cleaning.h); to a large extent this is just enforcing the Slavic consonant-cluster restrictions discussed in Section 1.5.

The inflection-tables themselves are indexed by integer-keys associated with each inflection-class, such that the 'correct' table's key always ends on '1', meaning that shifting to the alternative tables is a matter of simply incrementing the key by either 1 or 2, and I can keep track of whether or not a form has required a deviant or alternative ending by inspecting the final value of this key<sup>26</sup>:

<sup>25</sup> Extra handling is required for forms of *\*byti*, since that has a separate future-paradigm (and future-participle) which TOROT does actually specify separately with an 'f' value for the *tense* feature, as well as two variant imperfect sets (3sg. *\*bě* vs. *\*běaše*, which aren't tagged, but which I detect), and a 'conditional' *\*bimъ*, *\*bi*, *\*bq* etc. Participles require an extra step to read their nominal-features and add their adjective-like endings.

<sup>26</sup> This is bad design and prevents me from easily handling more than one type of morphological innovation per class: for the *masc\_jo* class nom. pl. I have the i-stem *\*-ъje* ending in the 'deviances' table (as found in e.g. Supr. *стражѣ*

```
std::unordered_map<int, inner_map> verb_ = {
    {111, v_11_c0},
    {112, v_11_c1},
    {113, v_11_c2},
```

Figure 3: Integer-keys of the three inflection-tables in Figure 2

Multiple inflection-classes can share the same tables, for instance the *masc\_o*, *nt\_o*, and *fem\_a* noun-classes all take endings from the *adj\_hard* table, since the same set of 63 endings (21 for each gender) is used in all four paradigms<sup>27</sup>.

Full paradigms for the 4,500-ish OCS lemmas I have so far reconstructed can be dynamically generated here <https://ocstexts.co.uk/words>; these are constructed from LCS lemmas in the same way the Autoreconstructor reconstructs individual forms, except it produces every form in the paradigm, rather than just the one specified by a text-word's morphology-tag. The LCS forms are then converted into 'normalised' OCS by the browser using the `convertToOCS()` function here: [https://github.com/12401453/ocs\\_server/blob/main/HTML\\_DOCS/LCS\\_to\\_OCS.js#L197](https://github.com/12401453/ocs_server/blob/main/HTML_DOCS/LCS_to_OCS.js#L197). Forms generated from the "deviant" and "alternative" tables are included, but the former are given less prominent styling to identify them as (probable) post-LCS innovations.

<\*storž-ɣje ‘guards’), which leaves no room for forms like Supr. 3H016E <\*znoj-eve, Psal. 0ᠠᠨᠵᠡᠨᠡᠭᠡᠨᠡᠭᠡ {zmieve} <\*zmɣj-eve, which have the \*-eve ending from the ju-stems.

27 The long-form adjectives (and participles) are then formed by just sticking the inflected-form of the pronoun \*jъ onto the end, since this is the origin of such long-forms (Vaillant 1942). I agree with Townsend and Janda (1996: 178) that some simplification of these concatenated endings must have occurred by LCS, especially where the short-form endings contain \*m or \*x (e.g. the dat. and loc. pl. for all genders), because it's unreasonable that e.g. Mar. Matt. 24 masc. loc. pl. **ⲛⲉⲃⲉⲥⲩⲕⲩⲭⲓⲭⲓ** {nebesūsķūixū} could've developed directly from \*nebesъskēxъjixъ (here not least because of the lack of PV2-reflex). I am however reluctant to adopt the (undiscussed) simplified LCS forms they present in the tables on p. 182-3 before I've done a more thorough investigation of early manuscript-forms, so for now I've left all such adjectives wholly uncontracted (though the Autoreconstructor does actually mark long-adjectivals that contain such problematic concatenations and this information could be used to exclude them from searches).

сѣти

Random

Corpus forms

|         | Present              | Aorist               | Imperfect                  | Imperative |
|---------|----------------------|----------------------|----------------------------|------------|
| 1st sg. | сѣтѣрѣ               | сѣтѣрѣхъ             | сѣтѣрѣахъ                  | сѣтѣрѣмь   |
| 2nd sg. | сѣтѣрѣши             | сѣтѣрѣ<br>сѣтѣрѣ     | сѣтѣрѣаше                  | сѣтѣри     |
| 3rd sg. | сѣтѣрѣтъ             | сѣтѣрѣ<br>сѣтѣрѣ     | сѣтѣрѣаше                  | сѣтѣри     |
| 1st du. | сѣтѣрѣвѣ             | сѣтѣрѣхѣвѣ           | сѣтѣрѣахѣвѣ                | сѣтѣрѣвѣ   |
| 2nd du. | сѣтѣрѣта             | сѣтѣрѣта             | сѣтѣрѣашета                | сѣтѣрѣта   |
| 3rd du. | сѣтѣрѣте<br>сѣтѣрѣта | сѣтѣрѣте<br>сѣтѣрѣта | сѣтѣрѣашете<br>сѣтѣрѣашета | сѣтѣрѣте   |
| 1st pl. | сѣтѣрѣмъ             | сѣтѣрѣхомъ           | сѣтѣрѣахомъ                | сѣтѣрѣмъ   |
| 2nd pl. | сѣтѣрѣте             | сѣтѣрѣте             | сѣтѣрѣашете                | сѣтѣрѣте   |
| 3rd pl. | сѣтѣрѣтъ             | сѣтѣрѣ<br>сѣтѣрѣша   | сѣтѣрѣахъ                  | сѣтѣрѣ     |

Participles

PRAP<sup>1</sup>

сѣтѣрѣи

PRAP<sup>2</sup>

сѣтѣрѣиѣ

PAP

сѣтѣрѣ

L-Part.

сѣтѣрѣаѣ

PPP

сѣтѣрѣтъ  
сѣтѣрѣиѣ

PrPP

сѣтѣромъ

Infinitive

сѣтѣрѣти

Supine

сѣтѣрѣтъ

Figure 4: Dynamically-generated paradigm for the OCS verb сѣтърѣти, based on the Autoreconstructor's computerised LCS inflectional-morphology

The accuracy of the generated-forms can then be gauged by comparing them to tables populated only by forms which actually occur in Eckhoff's corpus-texts, using the 'Corpus-forms' switch:

сѣти

Random

Corpus forms

|         | Present              | Aorist            | Imperfect | Imperative |
|---------|----------------------|-------------------|-----------|------------|
| 1st sg. |                      |                   |           |            |
| 2nd sg. |                      |                   |           |            |
| 3rd sg. | сѣтърѣтъ<br>сѣтърѣтъ | сѣтърь<br>сѣтърьє |           |            |
| 1st du. |                      |                   |           |            |
| 2nd du. |                      |                   |           |            |
| 3rd du. |                      |                   |           |            |
| 1st pl. |                      |                   |           | сѣтърьѣмъ  |
| 2nd pl. |                      |                   |           |            |
| 3rd pl. |                      | сѣтърьша          |           |            |

| Participles       |                      |
|-------------------|----------------------|
| PRAP <sup>1</sup> |                      |
| PRAP <sup>2</sup> |                      |
| PAP               | сѣтърьъ<br>сѣтърьъши |
| L-Part.           | сѣтърьаъ             |
| PPP               | сѣтърьени            |
| PrPP              |                      |
| Infinitive        |                      |
| Supine            | сѣтърътъ             |

Figure 5: Forms of the same сѣтърѣти verb as they actually occur in Eckhoff's (Cyrillicised) TOROT corpus of Church Slavic texts



An advantage of basing computer-generated paradigms on converted LCS (rather than directly on normalised OCS) is that it allows for the possibility of dialect or manuscript-specific conversions: e.g. Kiev Folia-flavoured paradigms that convert \*h, \*ĥ to <ц, з> and use <ѣ> for *all* \*Ē, or a Marianus-specific conversion that frequently shows vocalised strong-jers, or even an artificial pre-Jer Shift form of Old Russian that shows the denasalisation of \*ę and its merger with \*Ē via stochastic mixing up of <ѡ, ѡ> and <ѡ>:

|        | Sing.                | Dual                 | Plural                |
|--------|----------------------|----------------------|-----------------------|
| Nom.   | КЪНІАЗЬ<br>КЪНАГЪ    | КЪНАЗІА              | КЪНАЗІИ               |
| Acc.   | КЪНАЗЬ<br>КЪНАГЪ     | КЪНАЗА               | КЪНАЗѢ<br>КЪНАГЪІ     |
| Gen.   | КЪНАЗІА              | КЪНАЗЮ               | КЪНАЗЬ<br>КЪНАГЪ      |
| Dat.   | КЪНАЗЮ               | КЪНАЗѢМА<br>КЪНАГОМА | КЪНАЗѢМЪ<br>КЪНАГОМЪ  |
| Loc.   | КЪНАЗІИ<br>КЪНАЗѢ    | КЪНАЗЮ               | КЪНАЗІИХЪ<br>КЪНАЗѢХЪ |
| Instr. | КЪНАЗЬМЬ<br>КЪНАГЪМЬ | КЪНАЗѢМА<br>КЪНАГОМА | КЪНАЗІИ<br>КЪНАГЪІ    |
| Voc.   | КЪНАЖЕ               | КЪНАЗА               | КЪНАЗІИ               |

Figure 6: Dynamically-generated paradigm of \*кънѣзь but with an Old Russian rather than OCS conversion

|        | Sing.                | Dual                 | Plural               |
|--------|----------------------|----------------------|----------------------|
| Nom.   | КЪНАСЬ<br>КЪНАГЪ     | КЪНАСА               | КЪНАСИ               |
| Acc.   | КЪНАСЬ<br>КЪНАГЪ     | КЪНАСА               | КЪНАСА<br>КЪНАГЪІ    |
| Gen.   | КЪНАСА               | КЪНАСОУ              | КЪНАСЬ<br>КЪНАГЪ     |
| Dat.   | КЪНАСОУ              | КЪНАСѢМА<br>КЪНАГОМА | КЪНАСѢМЪ<br>КЪНАГОМЪ |
| Loc.   | КЪНАСИ<br>КЪНАСѢ     | КЪНАСОУ              | КЪНАСИХЪ<br>КЪНАСѢХЪ |
| Instr. | КЪНАСѢМЬ<br>КЪНАГОМЬ | КЪНАСѢМА<br>КЪНАГОМА | КЪНАСИ<br>КЪНАГЪІ    |
| Voc.   | КЪНАЖЕ               | КЪНАСА               | КЪНАСИ               |

Figure 7: \*кънѣзь with the canonical OCS conversion for comparison

## 2.3 Detecting and dealing with morphological innovations

|        |             |            |              |              |
|--------|-------------|------------|--------------|--------------|
| 217066 | приведоша   | 3paia----i | privedošę    | privedo      |
| 217101 | въздѣхнѣвъ  | -supamn-si | vъzdxnъvъ    | vъzdxъ       |
| 217112 | разврѣзосте | 3daia----i | orzvrъzoste  | orzvrъzete   |
| 217261 | бѣгивѣ      | -supamn-si | bolgoslovivъ | bolgoslovilъ |
| 217266 | ѣша         | 3paia----i | jĕšę         | jĕšę         |
| 217272 | възаша      | 3paia----i | vъzęšę       | vъzęšę       |
| 217302 | начаша      | 3paia----i | načęšę       | načęšę       |
| 217316 | въздѣхнѣвъ  | -supamn-si | vъzdxnъvъ    | vъzdxъ       |
| 217455 | приведоша   | 3paia----i | privedošę    | privedo      |
| 217600 | начать      | 3saia----i | načęť        | načę         |
| 217605 | сѣноу       | -s---md--i | synu         | synovi       |
| 217635 | начать      | 3saia----i | načęť        | načę         |
| 217787 | поѣтъ       | 3saia----i | pojęť        | poję         |
| 217832 | моѣомъ      | -s---mi--i | mosijomъ     | mosijemъ     |
| 217856 | моѣови      | -s---md--i | mosijovi     | mosiju       |
| 217869 | быс         | 3saia----i | bystъ        | by           |
| 217960 | сѣнѣ        | -s---ml--i | synę         | synu         |
| 218067 | невѣрьнѣ    | -s---mvpsi | nevęrъnъ     | nevęrъne     |
| 218108 | быс         | 3saia----i | bystъ        | by           |
| 218173 | нѣмѣ        | -s---mvpwi | nęmъjъ       | nęmęjъ       |
| 218175 | глоухѣ      | -s---mvpwi | gluxъjъ      | glušejъ      |
| 218198 | быстъ       | 3saia----i | bystъ        | by           |
| 218206 | оумрѣтъ     | 3saia----i | umerť        | umer         |
| 218388 | ѣмени       | -s---nl--i | jъmeni       | jъmene       |
| 218419 | ѣмени       | -s---nl--i | jъmeni       | jъmene       |
| 218561 | окомъ       | -s---ni--i | okomъ        | očesъmъ      |
| 218648 | моѣю        | -s---md--i | mōžu         | mōžęvi       |
| 218688 | моѣжа       | -s---mg--i | mōžĕ         | mōžu         |
| 218755 | прѣлѣубѣ    | -s---fa--i | perluby      | perlubъvъ    |
| 218762 | поустивѣши  | -supafn-si | pustivъši    | pušĕšĕi      |

Figure 8: Auto-detected and -reconstructed morphological deviances from a small part of the Book of Mark in Codex Zographensis

The screenshot above shows some raw data from my autoreconstructed SQLite database of the TOROT OCS texts; in this case it's forms from Zographensis (around Mark 7 to Mark 10) where the Autoreconstructor has detected morphological innovations. The fourth column shows what the Autoreconstructor thinks is the direct phonological ancestor to the text-form, but the ancestor of the 'original', 'correct', or 'default' morphological form is also generated and stored in the fifth column, so that such cases of innovation can be easily searched-for and counted (since non-innovated forms have NULL values in this column).

Types of innovation detected here include:

- extended S-aorists of class 1 verbs: 3<sup>rd</sup> pl. **приведоша** vs. **приведѣ**, 3<sup>rd</sup> dual **разврѣзосте** vs. **\*разврѣзете**<sup>28</sup>
- unetymological extension of the RUKI-rule-produced \*š in 3<sup>rd</sup> pl. primary sigmatic aorists: **ѣша** vs. **ѣса** <\*jĕd-s-ę, **възаша** vs. **възаса** <\*vъzym-s-ę, **начаша** vs. **начаса** <\*načъn-s-ę (neither \*d, \*m, nor \*n have ever been RUKI sounds)
- extension of the \*-nq- suffix to the past. act. part. of class 2 verbs like **въздѣхнѣти**: **въздѣхнѣвъ** (cf. Mar. **ⲱⲩⲗⲁⲩⲱⲩⲱⲩⲱⲩ** {usūxūšq} from **ⲱⲩⲗⲁⲩⲱⲩⲱⲩⲱⲩ**)

28 Koch (1990: 293) lists only sigmatic aorists as possibilities for the \*-verz-/\*-vřz- stem verbs, and it seems that outside of the 3<sup>rd</sup> sg. (e.g. Psal. **ⲕⲁⲩⲱⲩⲱⲩⲱⲩⲱⲩ** {razvrže}, Zogr. **ⲱⲩⲗⲁⲩⲱⲩⲱⲩⲱⲩ** {otvrže}) no root-aorists are attested in any Slavic text, so maybe I am wrong to set up asigmatic root-aorists like 3<sup>rd</sup> dual \*-vřzete as a possibility alongside primary sigmatic \*-verste (e.g. Mar. Mark 7 **ⲕⲁⲩⲱⲩⲱⲩⲱⲩⲱⲩ** {razvrže}). My justification is that the \*-verg-/\*-vřg- stem verbs do attest such root-aorists, e.g. 3<sup>rd</sup> pl. **ⲱⲩⲗⲁⲩⲱⲩⲱⲩⲱⲩ** {otvřręq} in Psal. Psalm 77, and I don't see what, apart from the nature of the final stem-consonant (obstruent vs. continuant), could be grounds for classifying these two verbs differently.

- addition of the \*-tQ suffix from the 3<sup>rd</sup> sg. pres. (where Q stands for NSI. \*Ъ vs SSI. \*Ь, see fn. 5 above) to 3<sup>rd</sup> sg. aorist forms: НАҢАТЪ, ПОІАТЪ, ОУМРЪТЪ
- original u/ju-stem nouns taking o/jo-stem endings: dat. sg. ѢНОУ, МЖЮ; loc. sg. ѢНѢ, gen. sg. МЖА
- past act. part. of class 4 verbs using the suffix \*-ivъ rather than \*-jъ: БГВІВЪ, ПОУСТИВЪШИ (cf. Mar. Mark 10 РѢШУОУШ {puštūši} <\*pust-jъši)

Deciding upon the “correct” morphological endings for an unattested language inevitably entails some uncertainty and controversy: for instance, \*-ox- aorists occur in both OCS and Old Russian, so why do I consider them LCS deviances? Basically because they **never**<sup>29</sup> occur in Marianus, which would be quite improbable if they were a discarded archaism (especially given their ubiquity in the closely related Zographensis). Additionally, West Slavic uses \*-ex- here (e.g. Old Czech 1sg. *nesech* < \*nes-ex-ъ), suggesting that the extended S-aorists were innovated independently in the post-LCS dialects (Meillet 1965: 256).

In other cases we are dealing with hodge-podge paradigms which are only ever attested with endings from multiple older classes, and sometimes dialectal or orthographic features of the manuscripts can make it difficult to distinguish potential LCS ancestors of certain endings. For instance, I have a *masc\_tel* class used for agent-nouns like OCS  $\Delta\tau\lambda\alpha\tau\epsilon\lambda\bar{\iota}$  (i.e. Diels 1963: 166), which in the sg. and dual. behave exactly like masc jo-stems, but which in the gen. and instr. plural are attested also with basically consonant-stem endings on a hard \*-tel- stem, e.g. Zogr.  $\text{тѣѣлѣ}$  {tŋʒatelŭ} <\*tež.Ātelŭ, Supr.  $\text{сѣдѣтелѣ}$  <\*sveĭtely. The nom. pl. appears also to take a consonant-stem \*-e ending, but as Diels (op. cit.) points out, the spellings in Zogr. and Supr. (where use of the palatalisation-diacritic <~> to denote LCS \*ĭ is consistent enough to suggest a real phonemic /ĭ/ in the underlying dialects) like  $\text{мѣстѣлѣ}$ <sup>30</sup> mean we can't follow Meillet (1965: 426) in setting up \*-tele as the 'correct' ending, because spellings like  $\text{жѣтеле}$  {žntele} in Mar. could just as easily descend from \*žetelē as from \*žetele. Absent a manuscript which both consistently marks \*ĭ and doesn't use such a mark in this nom. pl. desinence, there's no hard evidence of this \*-tele ending ever actually existing. I thus use \*-tele in the 'correct' table, and the jo-stem \*-telī in the 'deviances' table<sup>31</sup>.

## 2.4 Overcoming poor lemmatisation practice

Sometimes Eckhoff's lemmatisation is too coarse-grained, in that forms which clearly descend from distinct doublets are subsumed under one lemma. To demonstrate just one example of how the Autoreconstructor deals with this sort of problem, take the numeral *ѣдинъ* <\*jedinъ, which in the earliest OCS has straightforward hard pronominal endings and which therefore goes in my *pron\_hard* class alongside demonstratives like \*онъ. There is, however, what must be viewed as an LCS doublet \*jedьнъ<sup>32</sup>, which gives e.g. Serbian *jedan* and the modern Russian fem. nt. and

29 Having autoreconstructed all of Marianus I can verify this (admittedly already well-established) fact far more quickly and easily than was possible just with Eckhoff's morphology and part-of-speech annotation-information: using TOROT the smallest net you could cast would be one that caught all aorist-tense verbs, whereas I can just search for reflexes of \*oxъ, \*oxov, \*oxom, \*oše, \*osta\$, and \*oste\$ (the latter two using 'regular-expression' mode and \$ to specify end-of-word), which for Mar. turns up nothing except the three occurrences of ᾠστα. {osta}.

30 Searching my database for \*téle\$ returns only a single **ΒΛΑΨΤΕΛΕ** in Supr. without the diacritic; everything in Zogr. has it.

31 Diels mentions only Psal. Psalm 26 **𐩦𐩣𐩪𐩠𐩨𐩭𐩮** {*süvĕdētelj*}, but the Autoreconstructor is intended for use with other early texts beyond canonical OCS which might also contain this type of assimilation to the jo-stems.

32 This despite Meillet's (1965: 144) incoherent speculation about the /i/ in \*jedīnъ resulting from a phonetic development of strong \*ъ, analogous to what happens in the vicinity of \*j, because "*or il s'agit d'un composé dont le second élément est \*jīnū*". While this interpretation of \*jedīnъ's derivation could be true (and according to Derksen's (2008: 212) etymology of the \*jъnъ pronoun, its \*j is prothetic, meaning it wouldn't develop if already attached to a stem ending in \*d), the idea that a synchronic \*-dnъ by any purely phonological means could become /-din/, let alone early enough to completely displace by analogy the oblique forms in the earliest OCS where эднъ- {edin-} spellings are overwhelming, is ludicrous and contradicted by all the philological evidence.

oblique-case forms *одна, одного* etc., and which is used for the majority of non masc. nom./acc. sg. forms of the pronoun in Suprasliensis<sup>33</sup>, e.g. *ѦД'НОМОУ*. Notwithstanding Eckhoff's habit of lemmatising these with *ѦДИНЪ* instead of *ѦДЫНЪ*, it's trivial for the Autoreconstructor to check the form (which has in a previous step already been aggressively normalised to get rid of morphologically-irrelevant variation) for a <ДИН> sequence, which would never occur in the inflectional-ending and so always point to a \*jedin-descended stem, and then replace our autoreconstruction with \*jedьн- if such isn't found:

```
// check for *jedьн-
if (lemma_ref.lemma_id == compileTimeHashString("РѦДИНЪ") || lemma_ref.lemma_id == compileTimeHashString("МаѦДИНЪ"))
{
  if (Sniff(cyr_id, "дин", 20) == false)
  {
    stem = "jedьн";
  }
}
```

Figure 9: Part of the Autoreconstructor's code which checks for reflexes of \*jedьн- that TOROT has mistakenly lemmatised under *ѦДИНЪ*

In the long-term TOROT itself should fix such lemmatisation problems, but in the meantime checks like the above are computationally extremely cheap and prevent great swathes of the texts from being wrongly autoreconstructed.

### 3. Benefits and potential for future studies

We have already seen a few examples of minor questions where recourse to an LCS reconstructed layer makes exhaustive investigation of texts trivial: e.g. regarding the reality of a difference between word-initial /je-/ and (foreign) /e-/ in the Codex Suprasliensis shown by discrimination in use of the <Ѧ> letter (p.9-10), or the fate of the PV3 reflexes \*ś and \*з and whether they harden and merge with plain /s z/ or become palatalised counterparts to plain /s z/, i.e. /s' z'/, as in East Slavic (fn. 19).

LCS reconstructions are also useful for checking uncertain etymologies: if a text consistently spells a certain sound-group in securely-reconstructed words one way, but a different etymon which we less-assuredly reconstruct with that same sound-group is spelt differently, then that's a strong indication that our candidate-etymology is faulty. For example, the entry for Russian *бре́ние* in Vasmer's etymological dictionary (Фасмер 1986) mentions Sobolevskiy's suggested \*br̥nje reconstruction with syllabic \*r̥ instead of the more traditional \*br̥nje with a TR̥T-group, based on East Slavic *брьне* and *берне* spellings in a.o. the Izbornik 1073 and the Archangel'sk Gospel 1092. Sobolevskiy's view is that Russian *бре́ние* is a written borrowing from Church Slavonic which reflects the OCS <liquid + jer> ordering of the letters for syllabic liquids, viz. *брьне*. If we search for this lexeme in Psalterium Sinaiticum, however, we find three spellings: *БРЕНІѦ* {breniě}, *БРЕНІѦѦ* {breniě}, and *БРѦНІѦ* {br̥niě}, two of which suggest a regular vocalised /e/ reflex of strong \*br̥ in a TR̥T-group. Searching my autoreconstructed layer for \*r̥ turns up **zero** similar <бр̥> spellings in the 410 reflexes of \*r̥ found: overwhelmingly this group is spelt <бр̥> in Psal. Conversely, if we use the regex [t̥r̥ps̥s̥dfgh̥h̥kl̥l̥z̥x̥č̥v̥n̥m̥ž̥]r̥[t̥r̥ps̥s̥dfgh̥h̥kl̥l̥z̥x̥č̥v̥n̥m̥ž̥]+[ь̥] to search for the strong Tr̥T-group suggested by the traditional \*br̥nje etymology, we find multiple <бр̥> spellings: *ПОСКРЕЖІЦЕТ҃҃* {poskrežīcetū} <\*poskr̥ž̥hetQ, *ВІСКРЕС҃҃* {vīskresū} <\*v̥skr̥ž̥hetQ, *ДѦБРѦХ҃҃* {d̥brexū} <\*d̥br̥x̥, and *ПОСКРЕЖІШТЕТ҃҃* {poskrežīšetū} <\*poskr̥ž̥hetQ. The evidence of Psal. is therefore unambiguously in favour of the \*br̥n- reconstruction: other reflexes of strong Tr̥T-groups show the same distribution of spellings as this etymon, while none of

33 According to my database there are 145 reflexes of \*jedьн- in this category vs 46 from \*jedin-, though all 107 reflexes of the masc. nom./acc. sg. are from \*jedinъ. Interestingly the Uspenskij Sbornik (at least the parts included in TOROT), in contrast to later Russian, appears to completely lack \*jedьн-, though all except the single *ОДННОН* are spelt with the OCS <Ѧ/> reflex of initial \*je-.





statistics broadly support Jagić's "Jer Umlaut" theory, but since he only gives rough percentages and we don't have the exact data he draws them from, we have no way of verifying it ourselves. Having each word indexed by its LCS phonemes allows all instances of a weak-jer to be searched for, whether the scribe has omitted them or not

or of the theory advanced by Totomanova (2014: 14-28) about a purported merger of the jers in the earliest Bulgarian after palatal and palatalised consonants, parallel to the better-known Middle Bulgarian merger of the nasals in the same environment.,

- Reference to Shevelov's Konsonant vor e, i in

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## Appendix 1: Glagolitic transliteration table

| Glagolitic | Latin | Commonly-used Cyrillic |
|------------|-------|------------------------|
| ⱁ          | a     | А                      |
| ⱂ          | b     | Б                      |
| ⱃ          | v     | В                      |
| ⱄ          | g     | Г                      |
| ⱅ          | d     | Д                      |
| ⱆ          | e     | Е                      |
| ⱇ          | ž     | Ж                      |
| ⱈ          | z     | З, Ѵ                   |
| ⱉ          | z     | З, Ѵ                   |
| ⱊ          | ï     | Ї, І                   |
| ⱋ          | l     | І                      |
| ⱌ          | i     | И                      |
| ⱍ          | ǵ     | Ѓ, Ѕ, Ї                |
| ⱎ          | k     | К                      |
| ⱏ          | l     | Л                      |
| ⱐ          | m     | М                      |
| ⱑ          | n     | Н                      |
| ⱒ          | o     | О                      |
| ⱓ          | p     | П                      |
| ⱔ          | r     | Р                      |
| ⱕ          | s     | С                      |
| ⱖ          | t     | Т                      |
| ⱗ          | ü     | У, Ү                   |
| ⱘ          | f     | Ф                      |
| ⱙ          | x     | Х                      |
| ⱚ          | ω     | Ѡ                      |
| ⱛ          | θ     | Ѳ                      |
| ⱜ          | c     | Ц                      |
| ⱝ          | č     | Ч                      |
| ⱞ          | š     | Ш                      |
| ⱟ          | ć     | Ѣ                      |
| Ⱡ          | ũ     | Ѥ                      |
| ⱡ          | ĩ     | Ь                      |
| Ɫ          | ě     | Ѧ                      |
| Ᵽ          | u     | Ѹ, ѹ                   |
| Ɽ          | ü     | Ю                      |
| ⱥ          | Q     | Ѡ                      |
| ⱦ          | Q̇    | Ѡ̇                     |
| Ⱨ          | e     | Ѣ, ѣ                   |
| ⱨ          | N     | Ѧ                      |
| Ⱪ          | Ṅ    | Ѧ̇                     |